BOSS — Python Prototype vs. Claude-built Version

Comparison as of 2026-09-22. Prototype: `V:\donttouch\Malkris\boss python` (Streamlit). Main project: `V:\donttouch\Malkris\boss-claude` (Java/Spring Boot + Angular).

Dimension	Python Prototype	BOSS — Claude-built
Tech stack	Single Python file, Streamlit	Java/Spring Boot backend + Angular frontend, separate services
Database	None — one flat `financial_data.json` file, overwritten every change	PostgreSQL with versioned Flyway migrations
Login / users	None — anyone who opens it sees/edits everything	JWT login, password hashing, per-user accounts
Multi-tenancy	No — one shared global state	Yes — isolated per company/workspace
Roles / permissions	None	TENANT_ADMIN, CEO, CFO, Manager, Employee, Viewer — each sees only what they're granted
Concurrent users	Breaks — two people/tabs overwrite each other's data	Designed for many simultaneous users per company
Data input	Manually typed: 6 numbers total (employees, avg CTC, clients, price, contract, cash)	Real CSV/Excel transaction upload + 27 structured domain-specific forms with their own bulk upload
Historical data / trends	None — only "right now"; every save overwrites the last	Real multi-month history accumulates from every upload; trend chart, month-over-month comparison
Financial calculations	Inline arithmetic in the script (payroll, burn, revenue, profit, runway)	Dedicated tested calculators (Revenue/Expense/Profit/Margin/Burn/Runway), unit-tested
Forecasting	Flat 5-month projection, hardcoded month labels (Jan–May regardless of real date)	Real linear-regression (OLS) forecast over actual history, 3/6/12/24/36-month horizons
Scenario simulation	Recomputes the same live numbers as you move sliders — no saved base to compare against	Real base-vs-scenario comparison reusing the same forecast engine; revenue%, expense%, hiring inputs
Risk detection	5 hardcoded `if` thresholds	Rule-based RiskEngine over real snapshot history, evidence-labeled
Recommendations	Not separate from the risk list — just a warning line	Persisted, evidence-based recommendations (root cause, alternatives, confidence, follow-up)
Action tracking / outcomes	None	Full lifecycle: propose → approve → complete, with actual-vs-expected variance recorded
AI layer	Misnamed "AI CFO" — literally 5 `if` statements, no chat	Orchestrated chat grounded in real computed data (mock LLM by default, swappable)
Other business domains	Finance only	HR, Accounting, Auditing, Compliance, Risk register — 27 subcategories with real forms
Bulk upload	None	Yes, for every one of the 27 domain forms, validated row-by-row
Reporting / export	One button → 3-line PDF	None yet (a genuine gap)
UI	Basic Streamlit default theme	Custom neumorphic light-mode design, collapsible sidebar

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Data safety on error	Corrupted file → code silently deletes it	Validation errors are rejected with a clear message; nothing is silently destroyed
Tests	None	123+ backend tests, Playwright e2e suite
Deployment	`streamlit run` on one machine	Docker Compose (Postgres, Redis, backend, frontend)
Setup time	Seconds	Minutes (Docker images, migrations)
Best for	One person, one sitting, quick "what-if" guessing	A team running a real business, over months, with accountability

Where the Python version actually wins

Only one place, honestly: **PDF export**. It's a bare 3-line PDF, but it exists — BOSS has no export/reporting feature at all yet. Beyond that, and pure setup speed, there's no dimension where the prototype comes out ahead.

Verdict

The prototype has no row in this table where it's usable by more than one person or survives past a single sitting. BOSS is the real product; the prototype's only remaining value is as a five-minute scratchpad for testing a new formula before it gets built properly into BOSS.